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# Findings and lessons from KVAC so far

Christiano · Jin · Semar · Marco · Oana · Callum

| Part | Class | Items |
|---|---|---|
| **A1** | Bounds without a derivation, with a missing term, or with the wrong relation | 6 |
| **A2** | Typos in the equations and the cross-references | 11 |
| **A3** | The scheme in the figure is not the scheme in the proof | 6 |
| **A4** | The overview statements disagree with the theorems the paper proves | 3 |
| **A5** | Proof steps missing as printed | 2 |
| **A6** | Decisions where the paper says less than a formalization needs | 33 |
| **B** | Lessons about formalizing the paper | 19 |
| **C** | Lessons about engineering and process | 7 |
| **D** | Lessons about the use of AI assistants | 5 |

Part A holds 28 findings about the paper and 33 decisions of the formalization. Parts B to D hold 31 lessons.

The report is docs/findings/FINDINGS_AND_LESSONS.md on main, PR 270, 31 October 2026.

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## Background. Security as a game

### A SECURITY PROPERTY IS A GAME

| ELEMENT | ROLE WITH RESPECT TO THE PROPERTY |
|---|---|
| **Challenger** | The honest parties the property protects. It also enforces the rules of the game. |
| **Adversary** | The attacker the property must withstand. The parties it controls and the oracles it queries fix the threat model. |
| **Two worlds** | The real world is the scheme as run. The simulated world meets the property by construction. Indistinguishable worlds carry the property over to the scheme. |
| **Advantage** | \|Pr[1 in the real game] − Pr[1 in the simulated game]\|. |
| **Property** | No efficient (polynomial time in λ, the security parameter that sets key and group size) adversary tells the two worlds apart, except with negligible advantage, i.e. advantage shrinks faster than the inverse of every polynomial in λ. |

### WHAT A PROOF BY GAMES ESTABLISHES

| ELEMENT | WHAT IT ESTABLISHES |
|---|---|
| **Statement** | For every adversary, the advantage against the scheme is at most a sum of advantages against the underlying cryptographic primitives. |
| **Reduction** | Each term comes from an adversary that runs the scheme adversary as a subroutine against one underlying cryptographic primitive. |
| **Validity** | Each reduction must play the security game of its primitive exactly as defined. If a reduction needs more than that game offers, the security of the primitive does not imply the security of the scheme. |
| **Conclusion** | An attack on the scheme yields an attack on an underlying cryptographic primitive. Negligible advantages against the primitives give a negligible scheme advantage. |

Shoup, Sequences of games, ePrint 2004/332. Bellare and Rogaway, Code-based game-playing proofs, EUROCRYPT 2006, ePrint 2004/331.

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## Theorem 5.8, anonymity of μCMZ

THE PROPERTY, DEFINITION 4.4 (P. 30)

| | |
|---|---|
| **Real world** | The honest user, with attributes m. |
| **Simulated world** | The simulator Sim = (Sim.I, Sim.P). It knows no attributes. |
| **Claim** | No efficient adversaries A and D tell the two worlds apart beyond the bound of Theorem 5.8 (p. 40). |
| **Meaning** | The messages of the user reveal about m only what the credential predicates state (p. 29). |

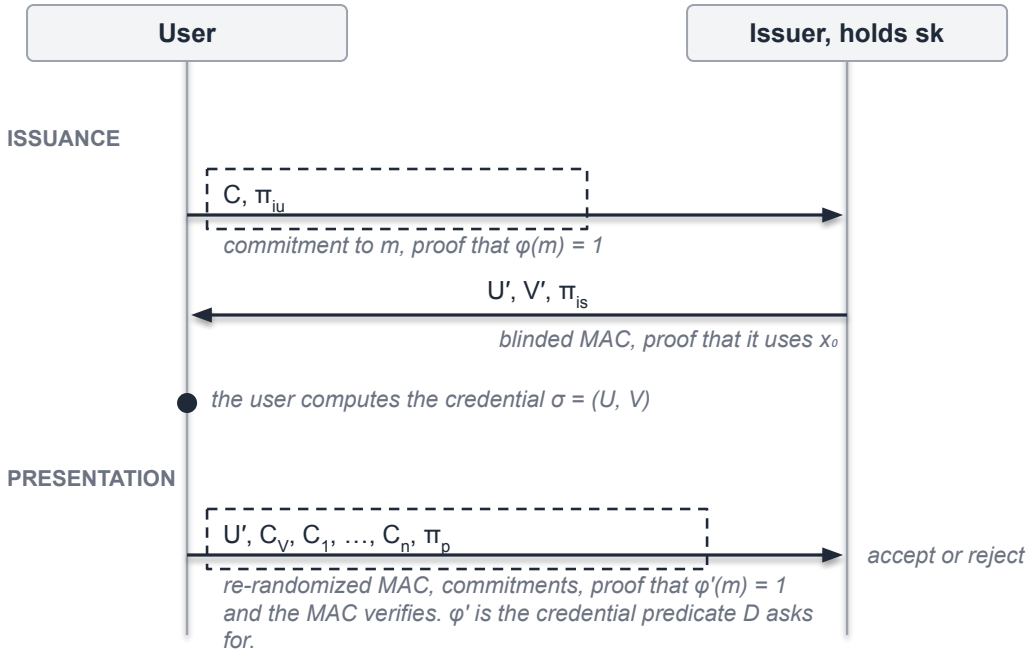
THE PART OF THE PROTOCOL, FIGURE 9 (P. 34)

| | |
|---|---|
| **Issuance** | The user sends C, $\pi_{iu}$. |
| **Presentation** | The user sends U′, $C_V$, $C_1$, …, $C_n$, $\pi_p$. |
| **Simulated** | Sim.I and Sim.P produce these framed messages. |

THE ADVERSARIES

| | |
|---|---|
| **A** | The issuer. It holds sk and may deviate from the protocol. |
| **D** | The verifier of presentations. It holds the state of A and chooses the credential predicates. |
| **Why sk** | Anonymity must hold with respect to the holder of the key. |

FIGURE 9 (P. 34)



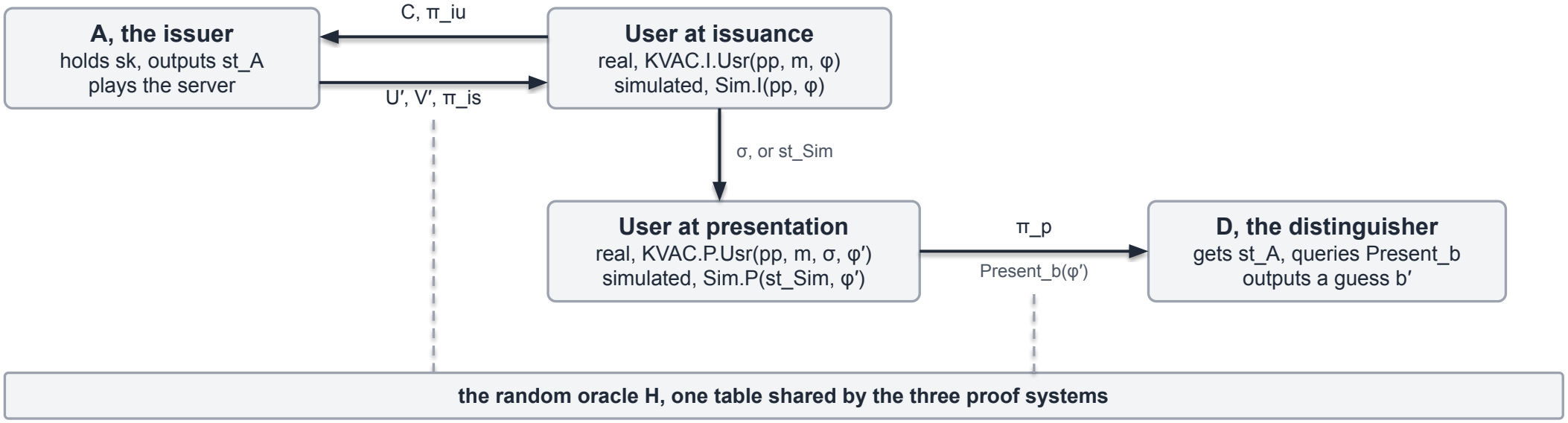

*Framed, the messages Sim.I and Sim.P replace. A plays the issuer. D receives the presentations.*

# Findings and lessons from KVAC so far

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**The anonymity game of Definition 4.4 (p. 30)**

*A plays the issuer, and D guesses whether the user runs the real protocol or the simulator.*




**Theorem 5.8 (p. 40).** If ZKP is a proof system for the relation $R \supseteq R_{cmz}$, then μCMZ is anonymous for $n = \text{poly}(\lambda)$ attributes with advantage $\text{Adv}^{\text{anon}}_{\mu\text{CMZ},A,D}(\lambda, n) \leq \text{Adv}^{\text{zk}}_{\text{ZKP_cmz.iu},A'}(\lambda) + \text{Adv}^{\text{zk}}_{\text{ZKP_cmz.p},D'}(\lambda) + \text{Adv}^{\text{ksnd}}_{\text{ZKP_cmz.is},A''}(\lambda)$, where A, D are adversaries in the anonymity game and A′, A″, D′ are described in the proof.

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## The proof of Theorem 5.8, step by step (pp. 40 to 41)

- A hybrid is a game of the proof. The real game is H0 and the simulated game is H5. Each hybrid changes one step of the previous one.
- With $p_i = \Pr[D \text{ outputs } 1 \text{ in } H_i]$, the triangle inequality ($|a - c| \le |a - b| + |b - c|$) gives $|p_0 - p_5| \le \Sigma_i |p_{i-1} - p_i|$.
- The left-hand side is the advantage of D (Definition 4.4). Each term $|p_{i-1} - p_i|$ is at most the cost of $H_i$ in the table. The sum of these costs is the bound of Theorem 5.8.

| HYBRID | CHANGE | COST IN THE BOUND | STATUS |
|---|---|---|---|
| **H0** | The real game. | | |
| **H1** | $\pi_{iu}$ from the zero-knowledge simulator, not from the witness. | $\mathrm{Adv}^{zk}_{ZKP_{cmz.iu},A'}(\lambda)$ | Holds. A′ plays the zero-knowledge game of $ZKP_{cmz.iu}$. |
| **H2** | $\pi_p$ from the zero-knowledge simulator. | $\mathrm{Adv}^{zk}_{ZKP_{cmz.p},D'}(\lambda)$ | Gap. D′ must also simulate $\pi_{iu}$, which programs the oracle H. The game of $ZKP_{cmz.p}$ owns the oracle H. |
| **H3** | $(x_0, u)$ extracted from $\pi_{is}$. | $\mathrm{Adv}^{ksnd}_{ZKP_{cmz.is},A''}(\lambda)$ | Gap. The extraction is from $\boldsymbol{\pi}_{is}$, but A saw a simulated $\boldsymbol{\pi}_{iu}$ first. The knowledge-soundness game of $ZKP_{cmz.is}$ (§3.3, p. 26) has no simulated $\pi_{iu}$. |
| **H4** | C random, MAC computed from the extracted witness. | 0, identical distribution | As printed. |
| **H5** | Presentation from $x_0$, with random U′, $C_V$, $C_i$. This is the simulated game. | 0, perfect hiding | Error. Z lacks $x_r U'$, and U′ must be nonzero (errata items 15 and 14). |

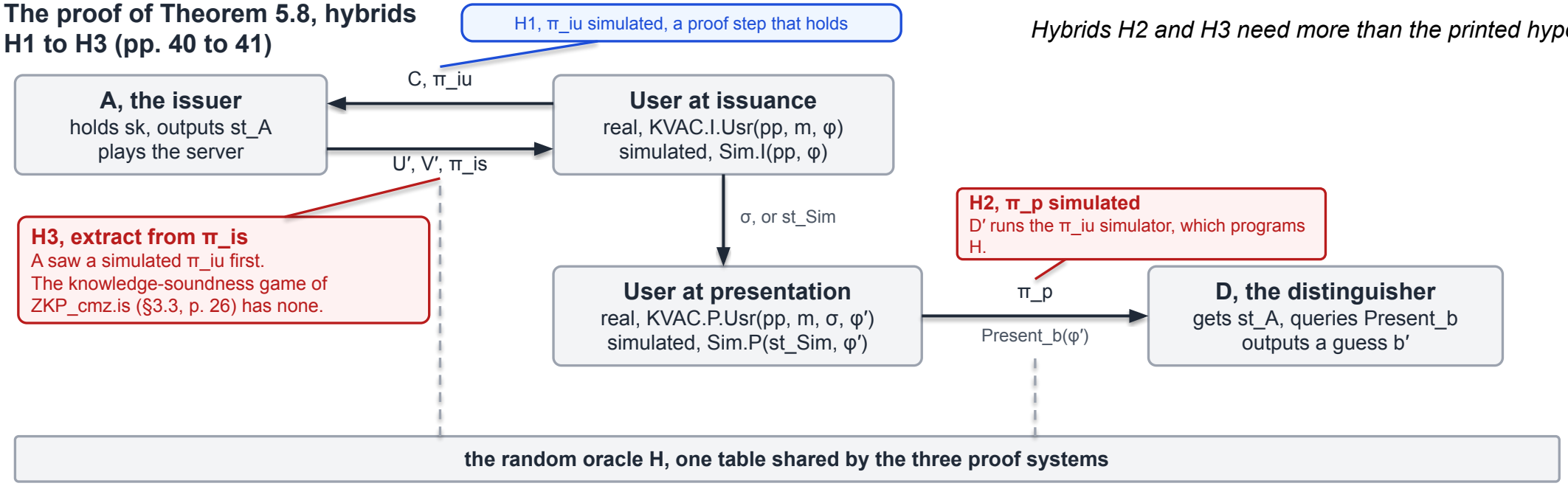
- A nonzero cost is the advantage of a reduction in the security game of a primitive.
- The reductions D′ of H2 and A″ of H3 need more than that game offers.

# Findings and lessons from KVAC so far

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## The proof of Theorem 5.8, hybrids H1 to H3 (pp. 40 to 41)

*Hybrids H2 and H3 need more than the printed hypothesis.*




**Theorem 5.8 (p. 40).** If ZKP is a proof system for the relation R ⊇ R_cmz, then μCMZ is anonymous for n = poly(λ) attributes with advantage Adv^anon_μCMZ,A,D(λ, n) ≤ Adv^zk_ZKP_cmz.iu,A′(λ) + Adv^zk_ZKP_cmz.p,D′(λ) + **Adv^ksnd_ZKP_cmz.is,A″(λ)**. The knowledge-soundness game of §3.3 behind the red term hands the prover no simulated proof, and H3 extracts after one. **The proof does not establish the theorem from the printed hypothesis alone.**

Item A4.1 of the Findings and lessons report

# Findings and lessons from KVAC so far

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## The fix

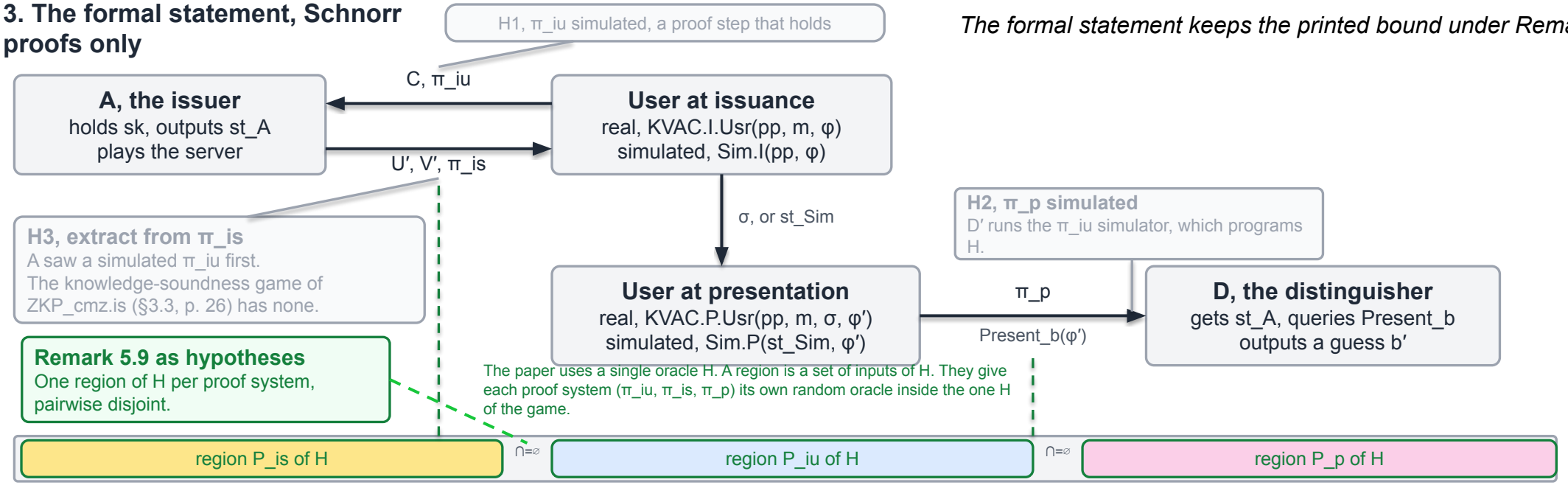
| | |
|---|---|
| **WHAT REMARK 5.9 (P. 42) STATES** | |
| **Requirement** | Simulated proofs of one type cannot be turned into proofs of another type. |
| **Case treated** | Extraction from $\pi_{iu}$ after a simulated $\pi_{is}$, the mirror of H3. |
| **General route** | Simulation-extractable proofs on different setup strings, or on **different random oracles**. |
| **Schnorr proofs** | Knowledge soundness suffices because the vector sizes of the Schnorr proofs of $R_{cmz.iu}$ and $R_{cmz.is}$ mismatch. No proof given. |
| **WHAT THEOREM μCMZ_anonymity ADDS** | |
| **Regions** | $P_{iu}$, $P_{is}$, $P_p$, sets of inputs of H. They implement the different random oracles of Remark 5.9 as regions of the one oracle H of the game of Theorem 5.8. |
| **Confinement** | Each proof system with its simulator or extractor reads and writes H inside its region only. |
| **Disjointness** | $P_{iu} \cap P_{is} = P_{iu} \cap P_p = P_{is} \cap P_p = \varnothing$. |
| **WHY THE GAPS CLOSE** | |
| **H3** | A″ runs the $\pi_{iu}$ simulator on $P_{iu}$. The extractor reads $P_{is}$ only, so the knowledge-soundness advantage applies. |
| **H2** | D′ programs H on $P_{iu}$ only, outside $P_p$, so the zero-knowledge game of $ZKP_{cmz.p}$ applies. |
| **Bound** | Unchanged. The hypotheses remain to be discharged for Schnorr proofs. |

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## 3. The formal statement, Schnorr proofs only

*The formal statement keeps the printed bound under Remark 5.9.*




**μCMZ_anonymity.** If ZKP is a proof system for the relation R ⊇ R_cmz**, with ZKP_cmz.iu, ZKP_cmz.is and ZKP_cmz.p, their simulators and the extractor confined to pairwise disjoint regions of the random-oracle domain**, then μCMZ is anonymous with advantage Adv^anon_μCMZ,A,D(λ, n) ≤ Adv^zk_ZKP_cmz.iu,A′(λ) + Adv^zk_ZKP_cmz.p,D′(λ) + **Adv^ksnd_ZKP_cmz.is,A″(λ).**

The proof of Theorem 5.8 needs the condition of Remark 5.9 (p. 42).